

$$b) E(P_{\text{normal}}) = 5 \cdot 24 = 120$$

$$\text{rate } \lambda = 3$$

$$= 3 \cdot 4 \text{ hrs} = 12 \text{ customers}$$

M = number of customers in 'new day'

$$P_{\text{new}} = 5 \cdot M$$

$$P(P_{\text{new}} \geq 120 | P_{\text{old}}) = P(L \geq 24)$$

The probability the profit on this new schedule is equal to the probability that the bakery gets more than 24 customers.

$$P(5M \geq 120)$$

$$P(M \geq 24)$$

$$\text{The opposite of } P(M \geq 24) = 1 - P(Y \leq 23)$$

$$P(\text{Profit} \geq 120) = 1 - F(23)$$

F is the cdf for the Poisson Random variable. $1 - F(23)$ is the chance that the amount of customers is greater than or equal to 24.